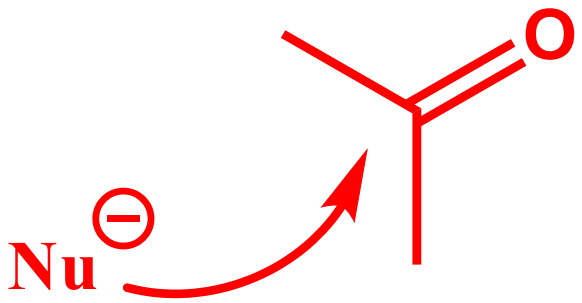
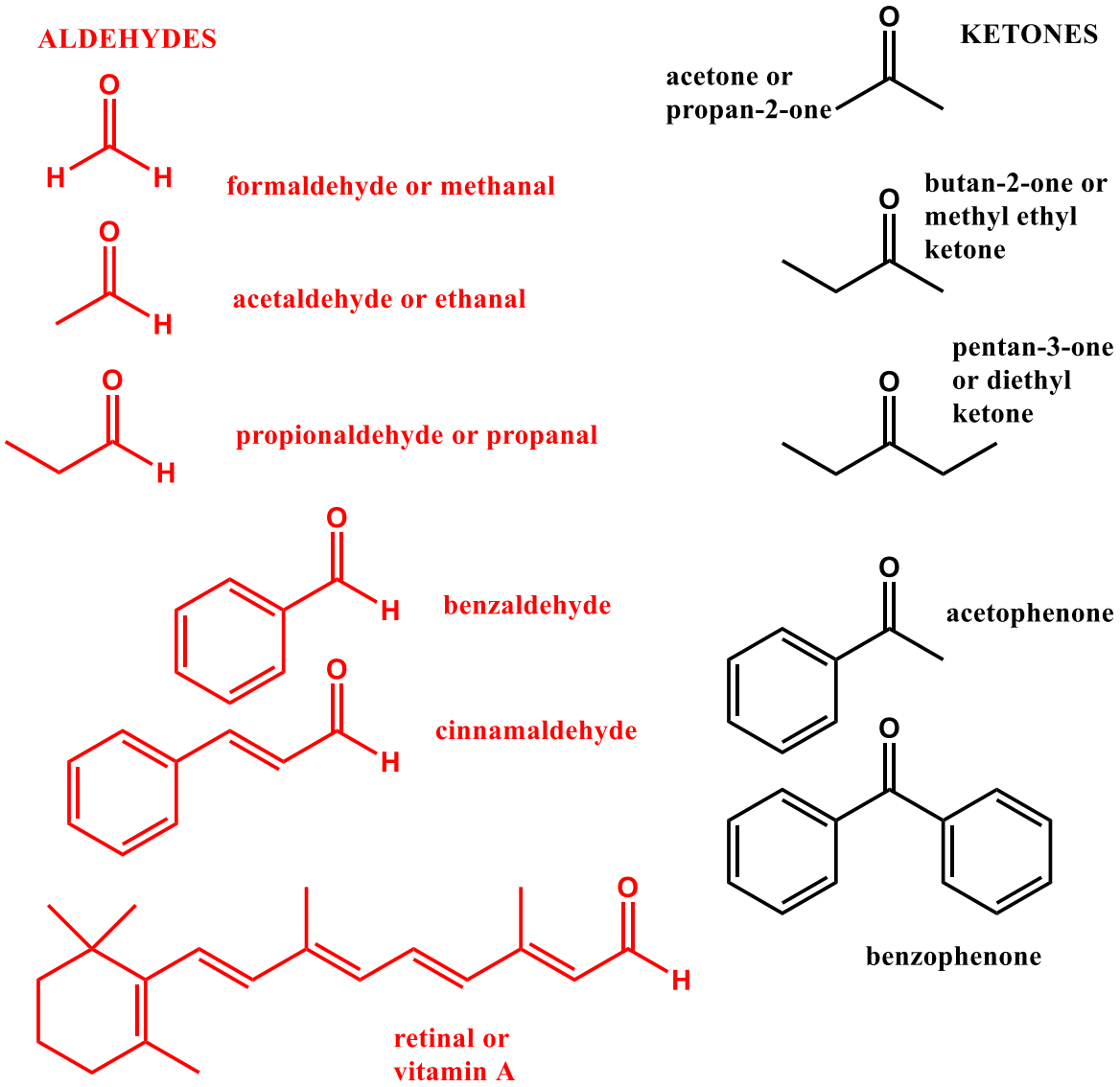
NUCLEOPHILIC ADDITION TO THE CARBONYL GROUP

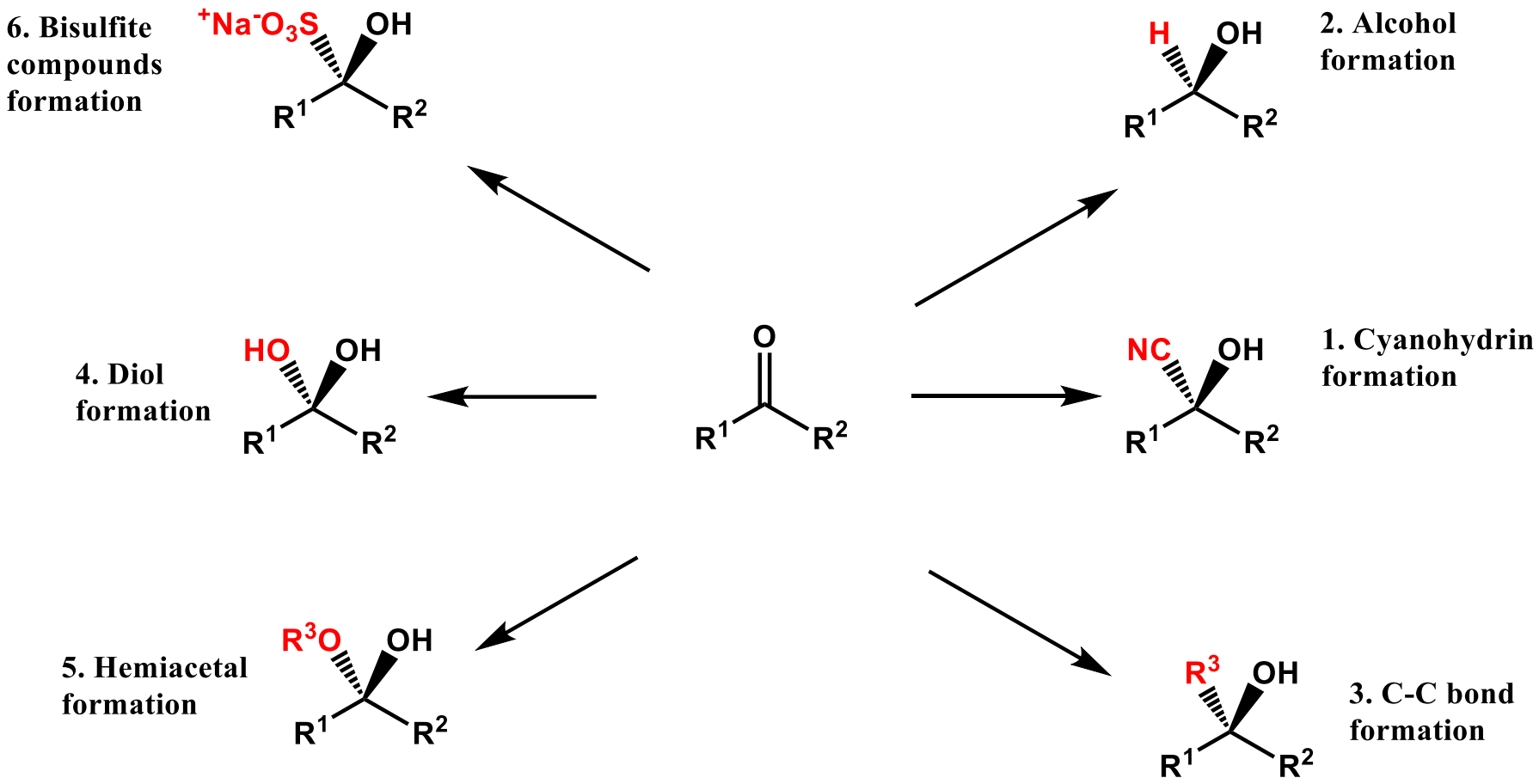


- How and why the C=O group reacts with nucleophiles
- Explaining the reactivity of the C=O group using molecular orbitals and curly arrows
- What sorts of molecules can be made by reactions of C=O groups
- How acid or base catalysts improve the reactivity of the C=O group

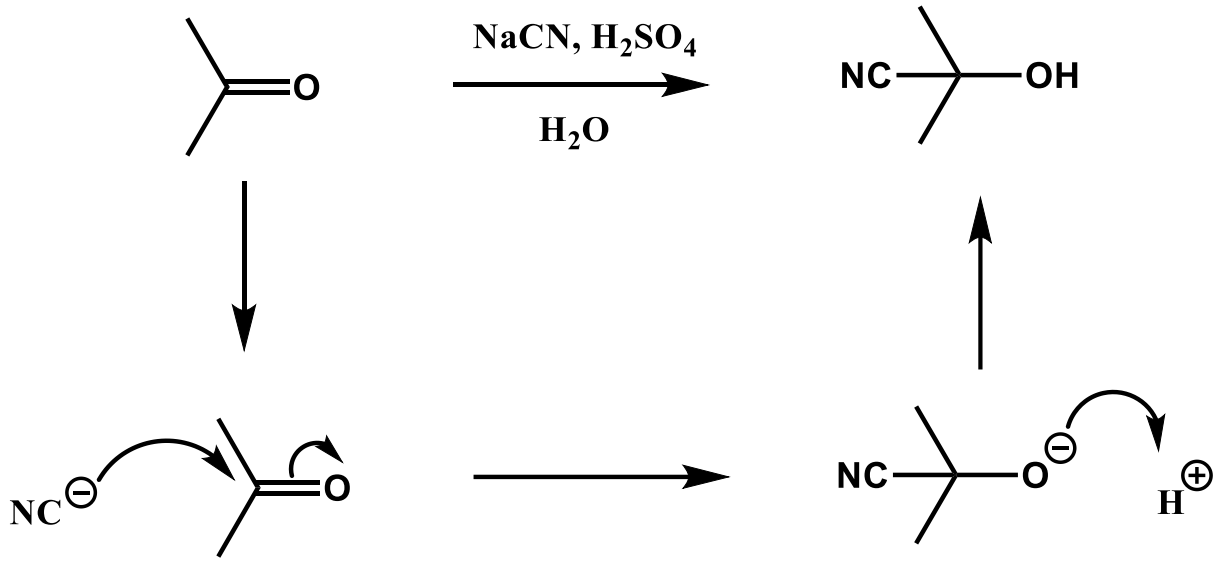
Carbonyl groups: Examples of Aldehydes and Ketones



What sorts of molecules can be made by reactions of C=O groups?



How does react the C=O group with nucleophiles?



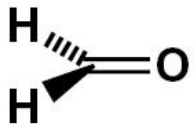
Nucleophilic addition of $^{-}$CN to The carbonyl group

Protonation

Additions to carbonyl groups generally consist of two mechanistic steps:

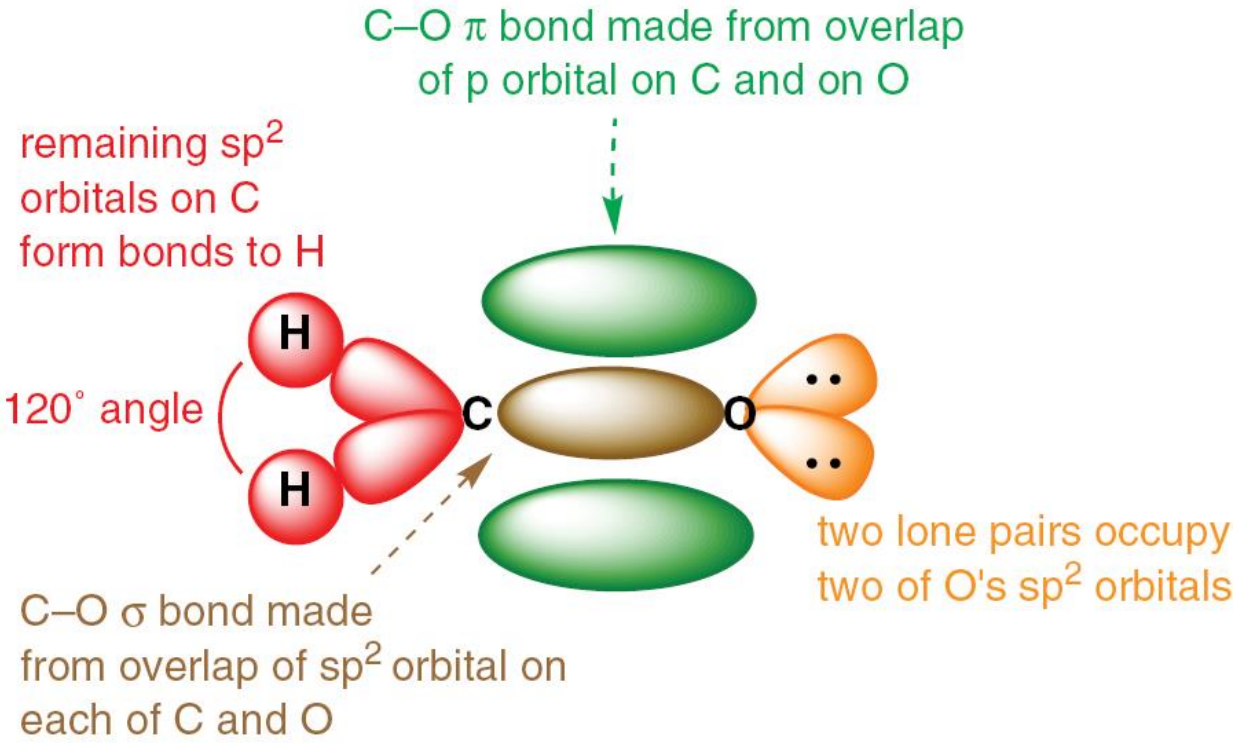
1. **Nucleophilic attack on the carbonyl group**
2. **Protonation of the anion that results**

Why the C=O group reacts with nucleophiles? Molecular Orbitals again!



formaldehyde
(methanal, CH_2O)

viewed from a slight angle



Why an electron-rich nucleophile is attracted by an electron-rich area, such as double bond???

Properties of the C=O group

Electronegativities, bond lengths, and bond strengths

Representative bond energies, kJ mol^{-1}				Representative bond lengths, Å				Electronegativity			
C–O	351	C=O	720	C–O	1.43	C=O	1.21	C	2.5	O	3.5

Two types of bonding between C and O:

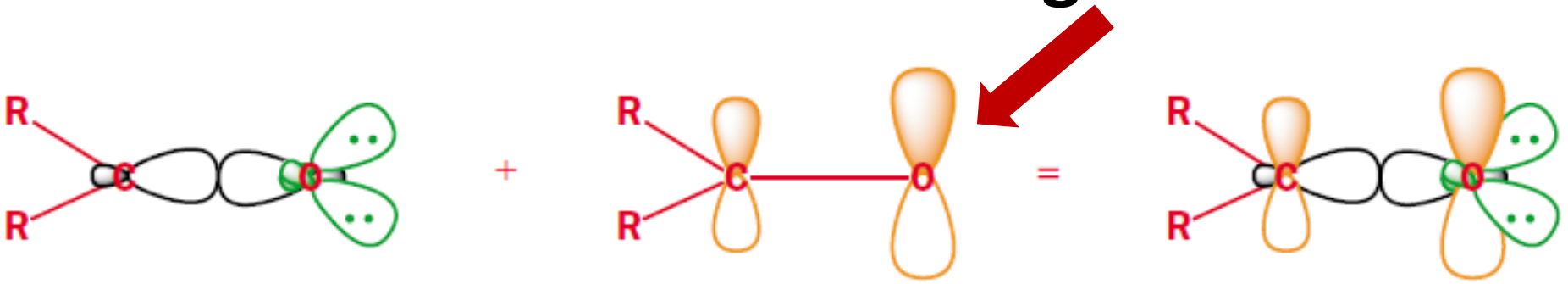
- **C=O double bond is rather shorter than a typical C–O single bond**
- **C=O more than twice as strong**

Why is it so reactive?

Polarization is the key

Carbon atom has some degree of positive charge and this charge attracts negatively charged nucleophiles

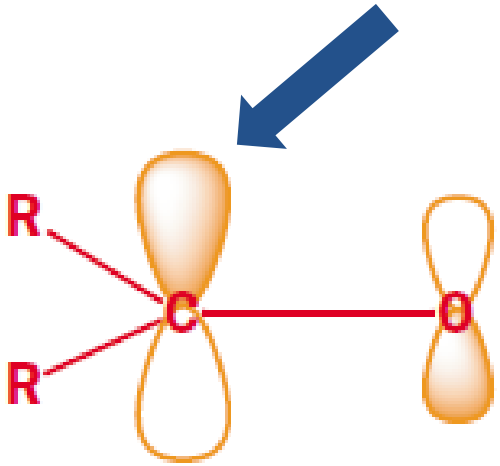
Why the C=O group reacts with nucleophiles? Molecular Orbitals again!



filled σ sp^2 (lone pair) orbitals

filled π orbital

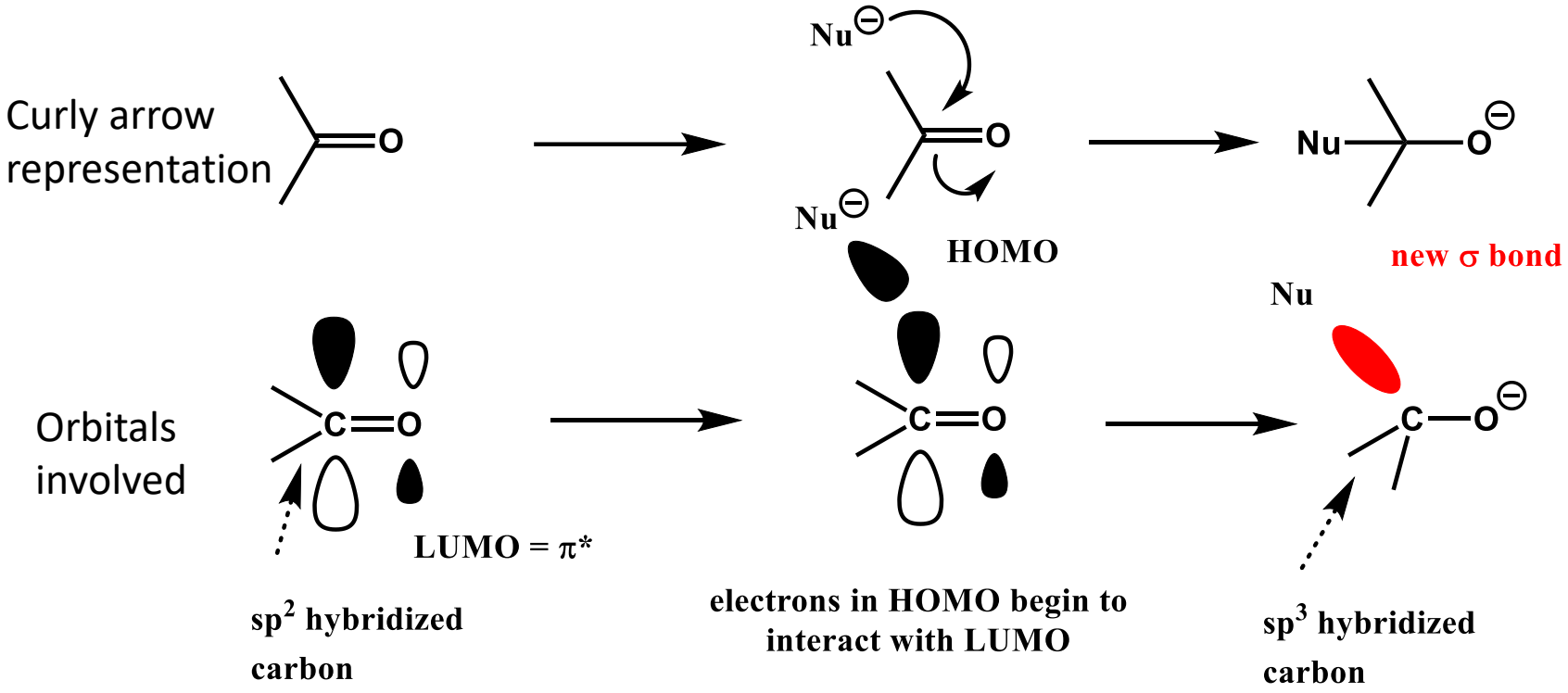
complete diagram of filled orbitals of C=O bond



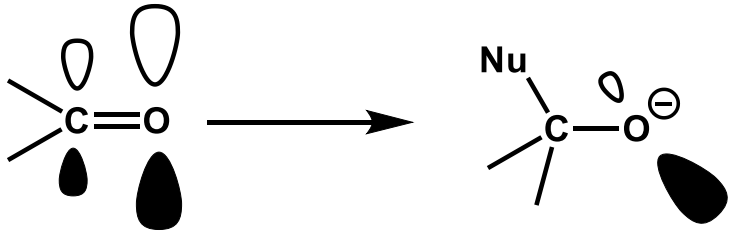
empty, antibonding π* orbital

This LUMO orbital will interact with the HOMO orbital of the Nu

How the C=O group reacts with nucleophiles? HOMO-LUMO interactions



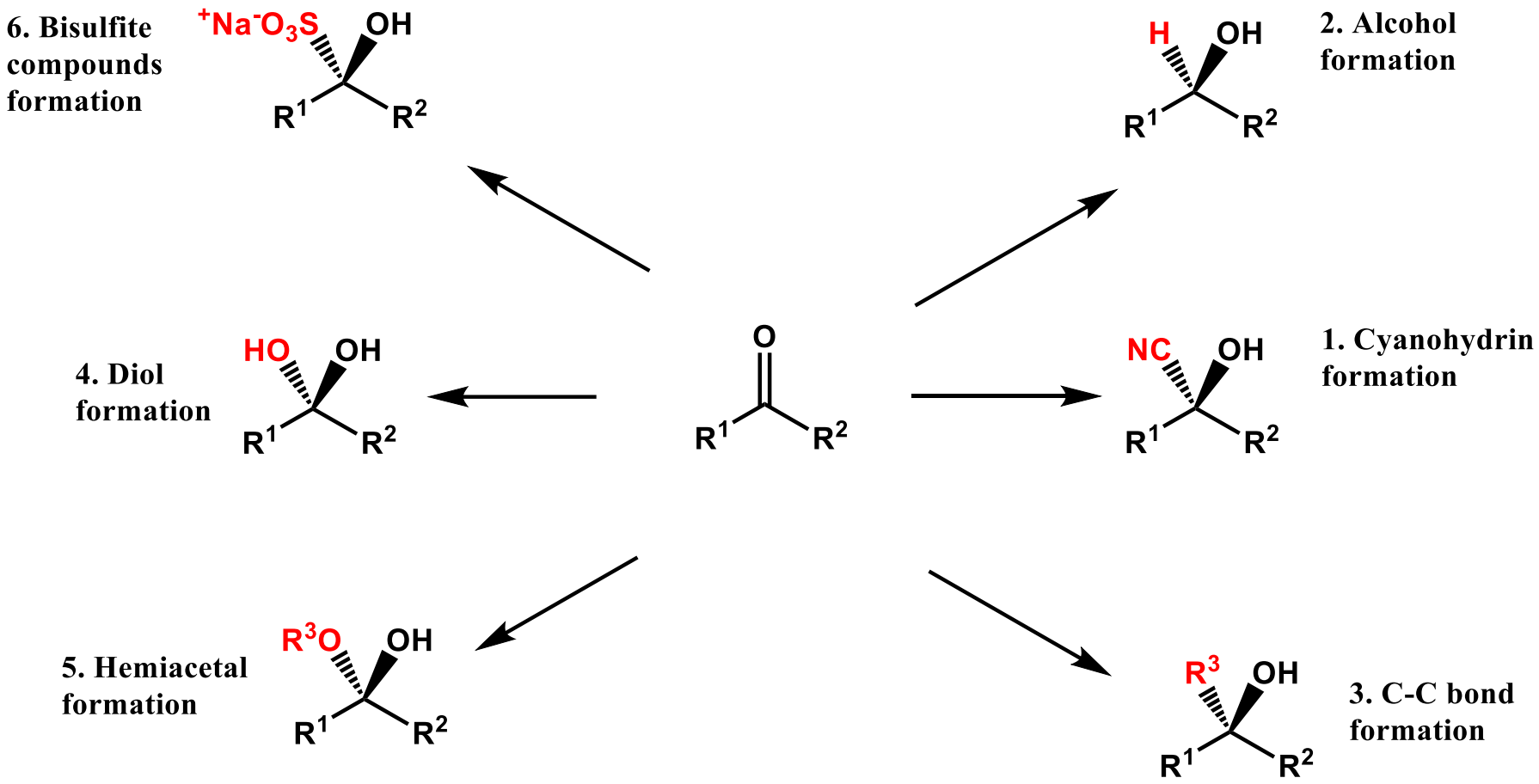
While at the same time



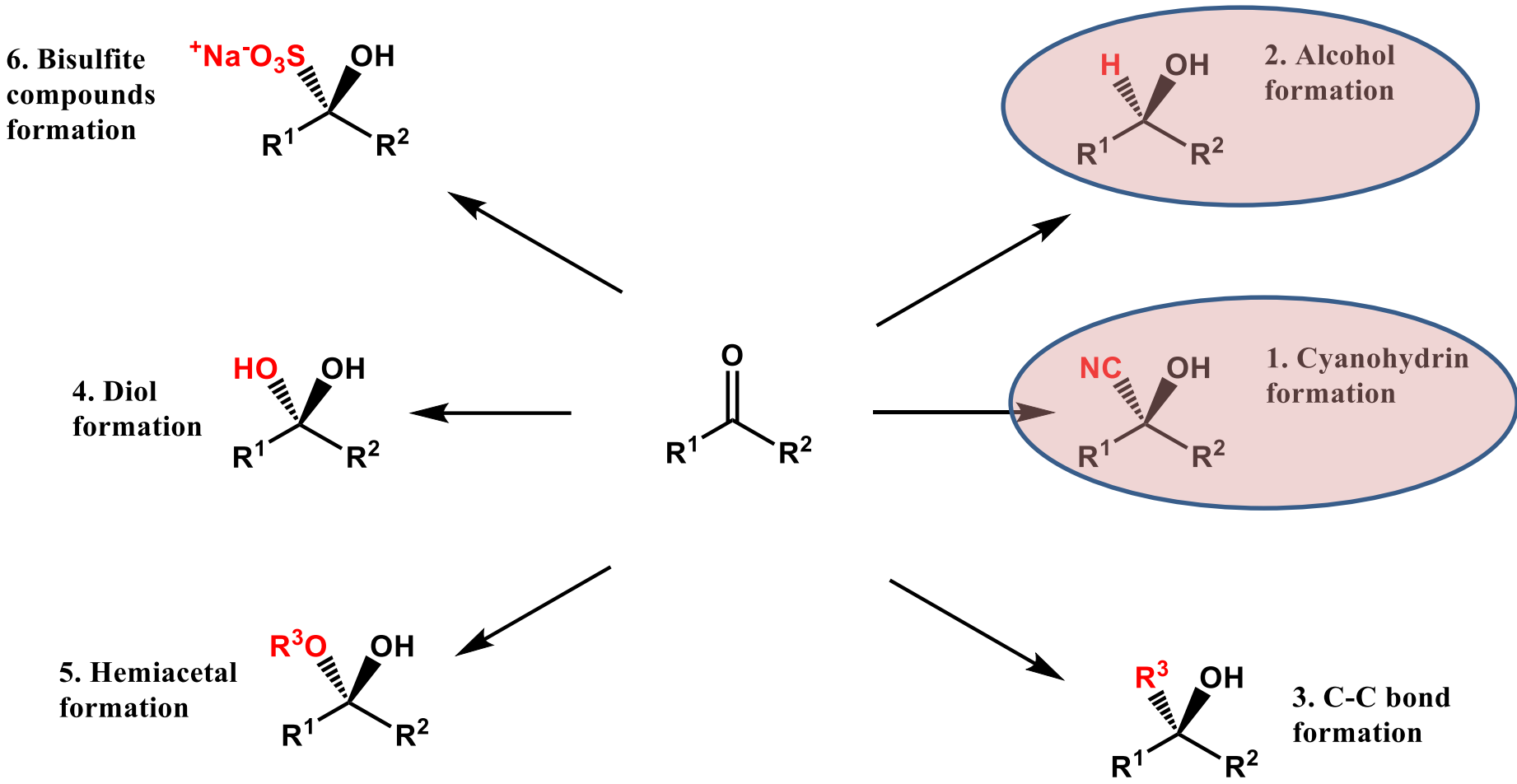
Nucleophilic addition to the C=O group: Summarizing…

- How and why the C=O group reacts with nucleophiles
- Explaining the reactivity of the C=O group using molecular orbitals and curly arrows
- **What sorts of molecules can be made by reactions of C=O groups**
- **How acid or base catalysts improve the reactivity of the C=O group**

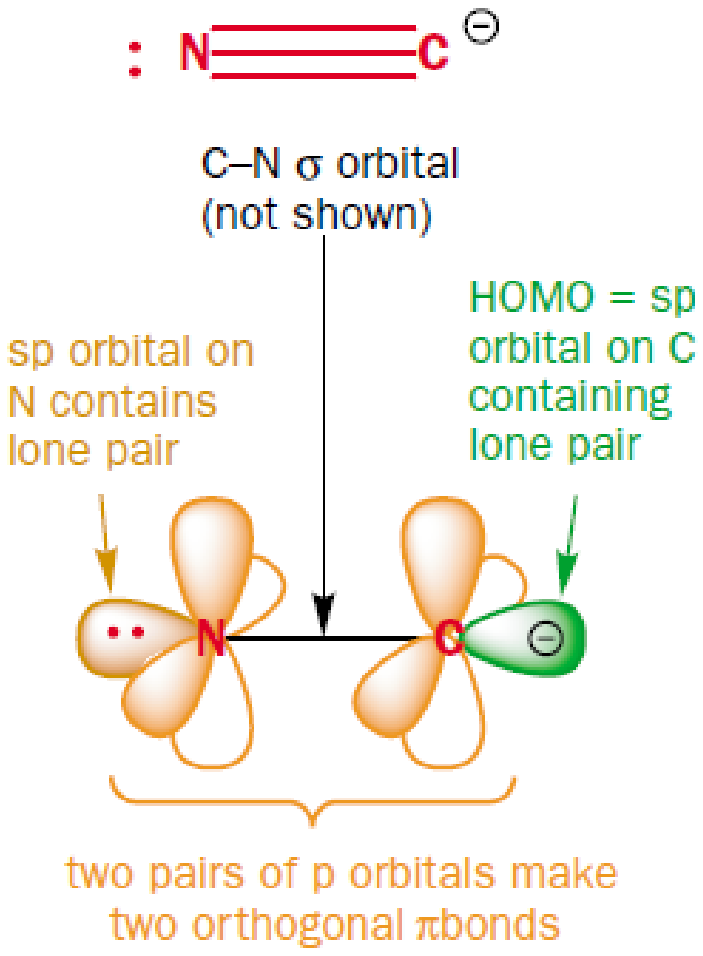
What sorts of molecules can be made by reactions of C=O groups?

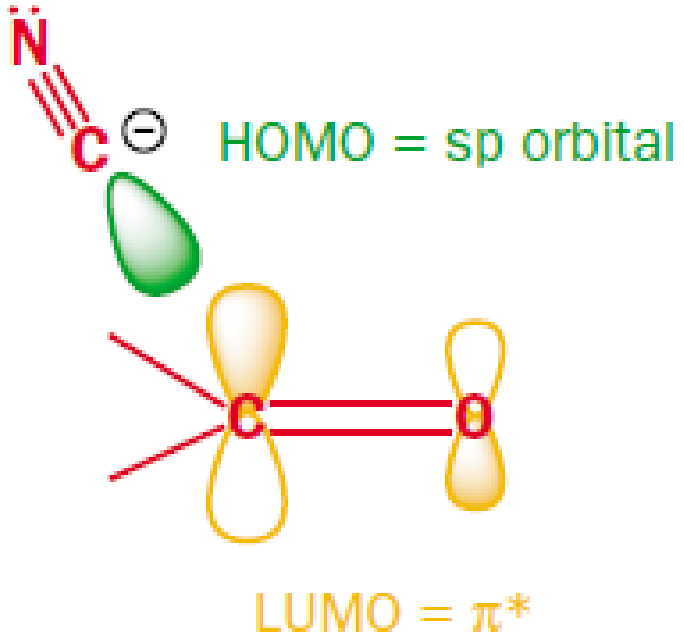


What sorts of molecules can be made by reactions of C=O groups?



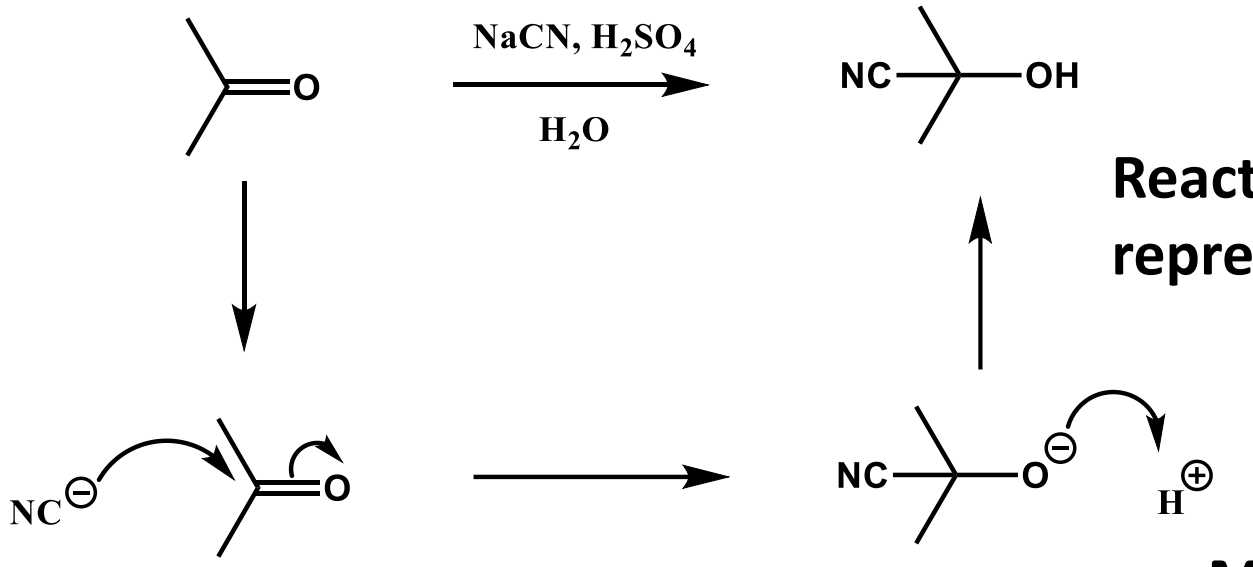
Cyanide addition: Cyanohydrin formation





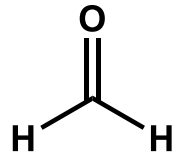
Cyanide addition: Orbitals involved

Cyanide addition: Cyanohydrin formation

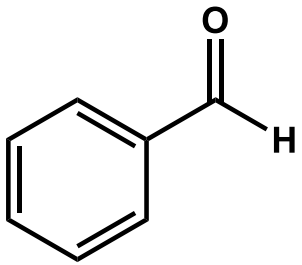


Cyanide addition: Cyanohydrin formation

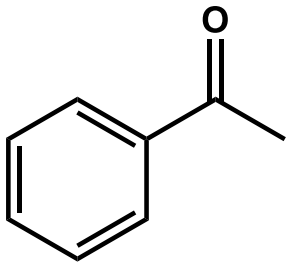
Complete these reactions:



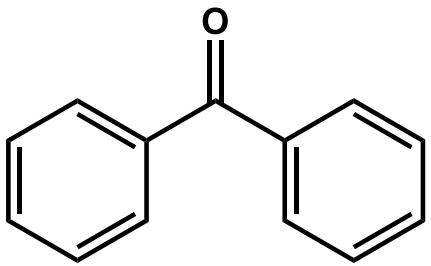
$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$



$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$

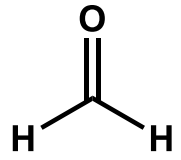


$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$

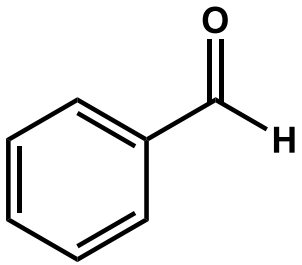


$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$

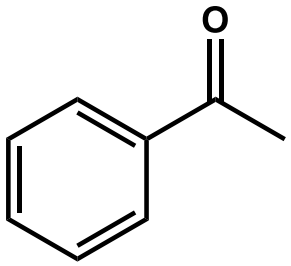
Which of these compounds is the most reactive?



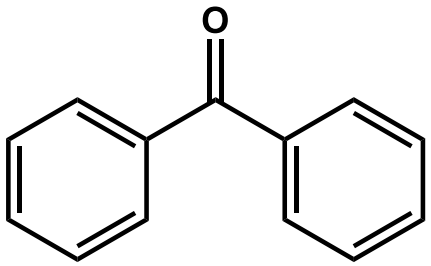
$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$



$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$

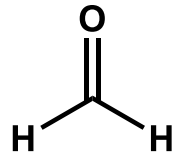


$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$



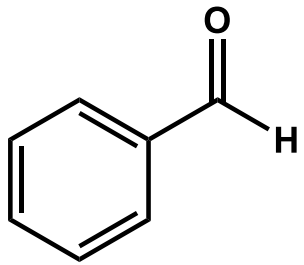
$\xrightarrow[H_2O]{NaCN,\ H_2SO_4}$

Which of these compounds is the most reactive?



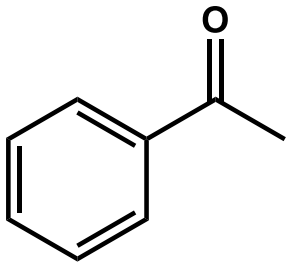
$NaCN, H_2SO_4$

H_2O



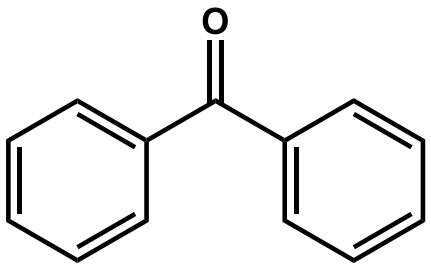
$NaCN, H_2SO_4$

H_2O



$NaCN, H_2SO_4$

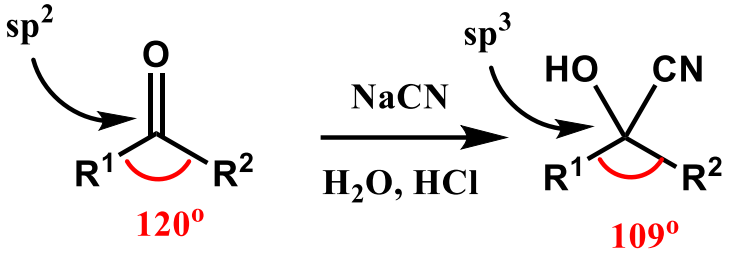
H_2O



$NaCN, H_2SO_4$

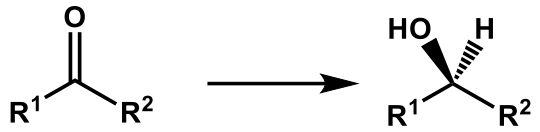
H_2O

Steric effects
Steric hindrance

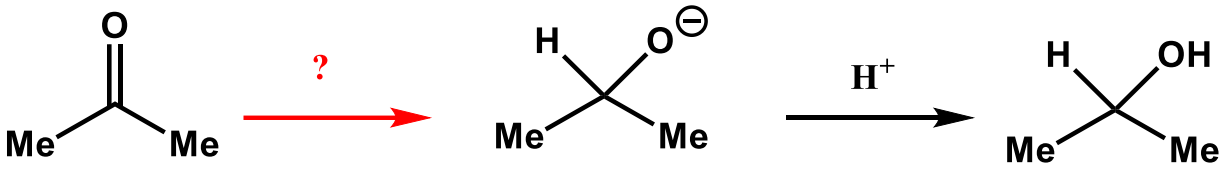


substituents move closer together

Alcohol formation from C=O compounds



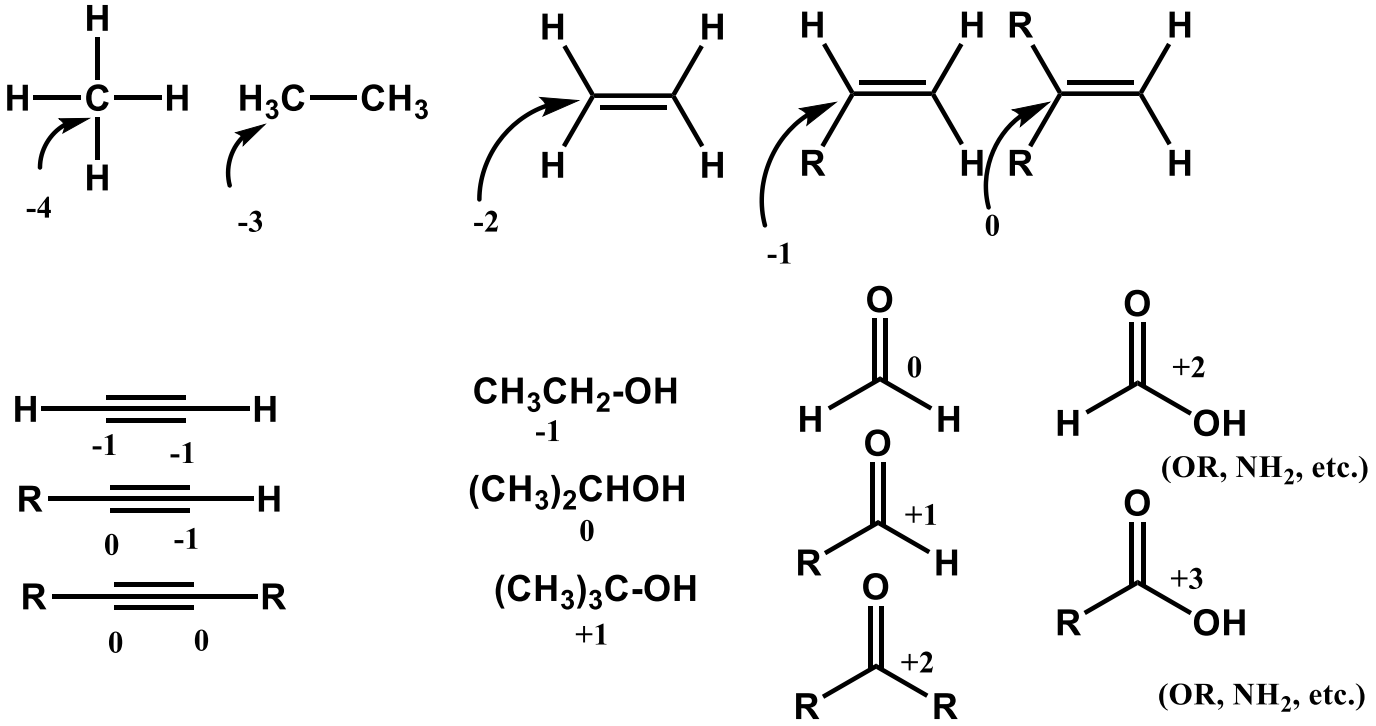
Reduction of a
Ketone to an alcohol



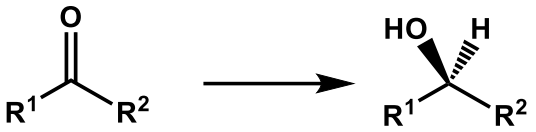
Oxidation levels

- **Bonds to hydrogen or other elements more electropositive than carbon contribute to -1 to the oxidation level.**
- **Bonds to other carbon atoms contribute 0 to the oxidation level.**
- **Bonds to oxygen or other elements more electronegative than carbon contribute +1 to the oxidation level.**
- **Multiple bounds to an element count as multiple single bonds to that element.**
- **A pair of electrons on carbon contribute -1 to the oxidation level.**
- **A positive charge on carbon contribute +1 to the oxidation level**

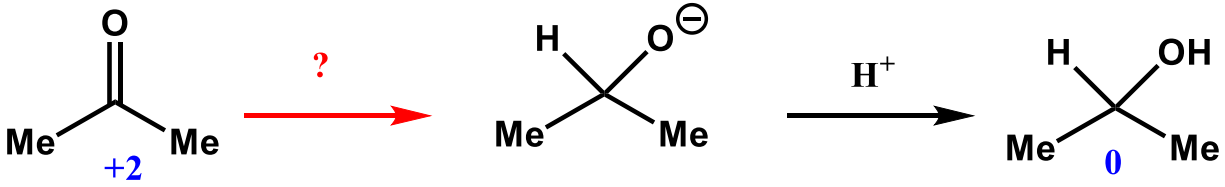
Oxidation levels

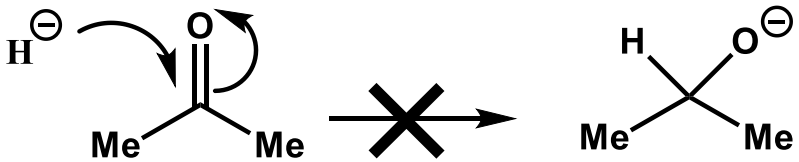


Alcohol formation from C=O compounds

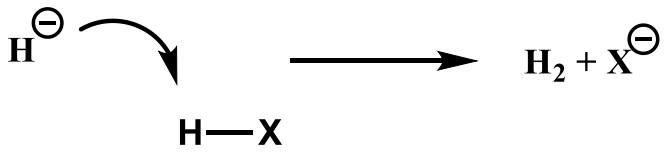


Reduction of a
Ketone to an alcohol



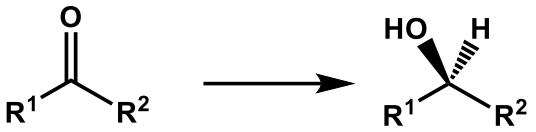


Nucleophilic attack by H^-
never happens

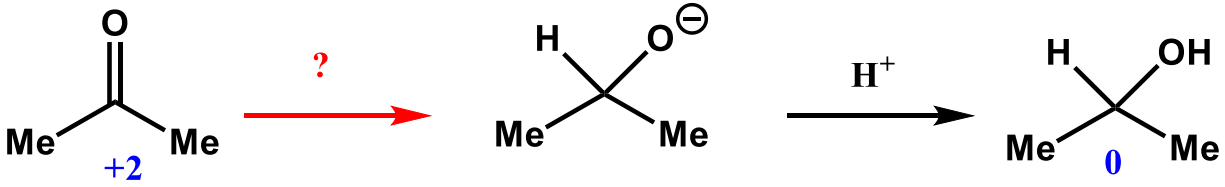


H^- always reacts as a base

Alcohol formation from C=O compounds

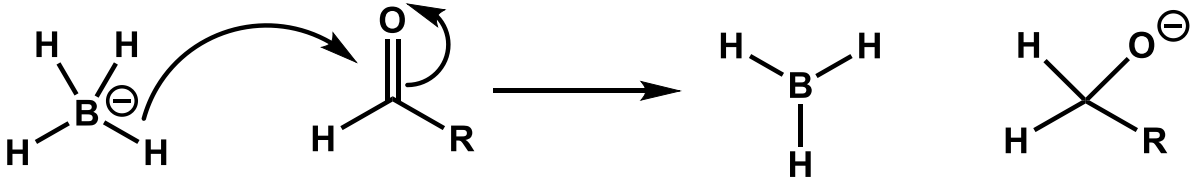


Reduction of a
Ketone to an alcohol

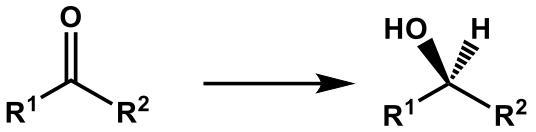


Where can we find nucleophilic H atoms?

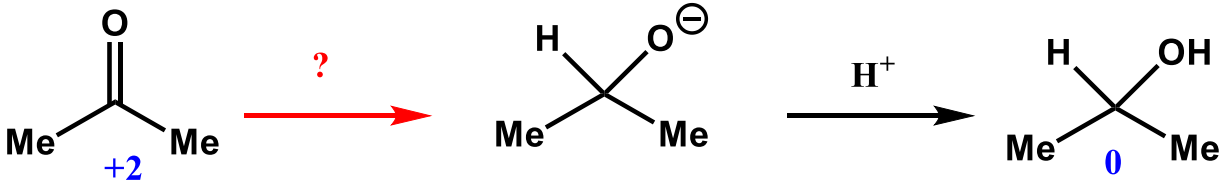
From sigma-bonds to electropositive atoms



Alcohol formation from C=O compounds

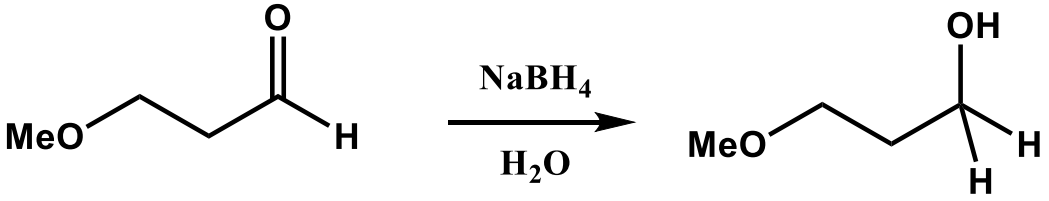


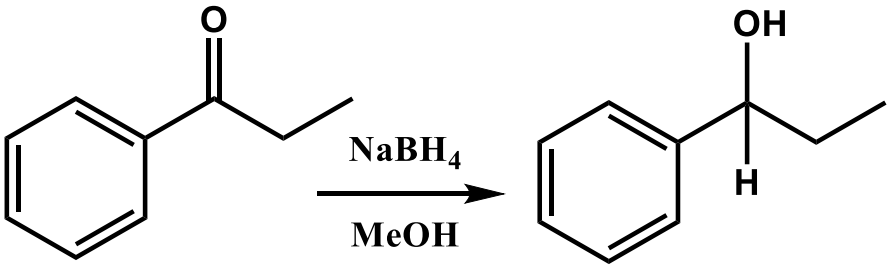
Reduction of a
Ketone to an alcohol

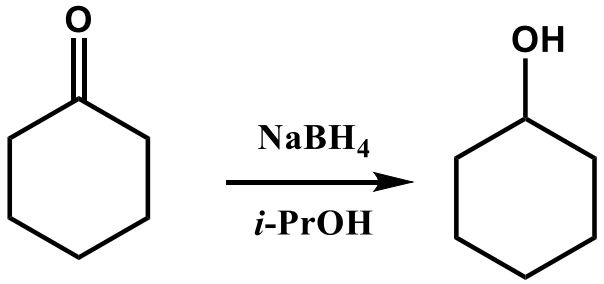


Where can we find nucleophilic H atoms?
From sigma-bonds to electropositive atoms

Examples of reductions
with sodium borohydride







C=O reduction by $NaBH_4$: Mechanism for the reaction

Arrow cannot start on negative charge: no lone pair on B

electrons must be transferred from a bond

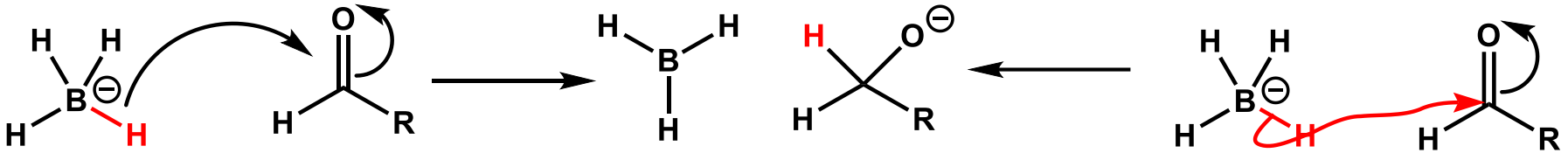


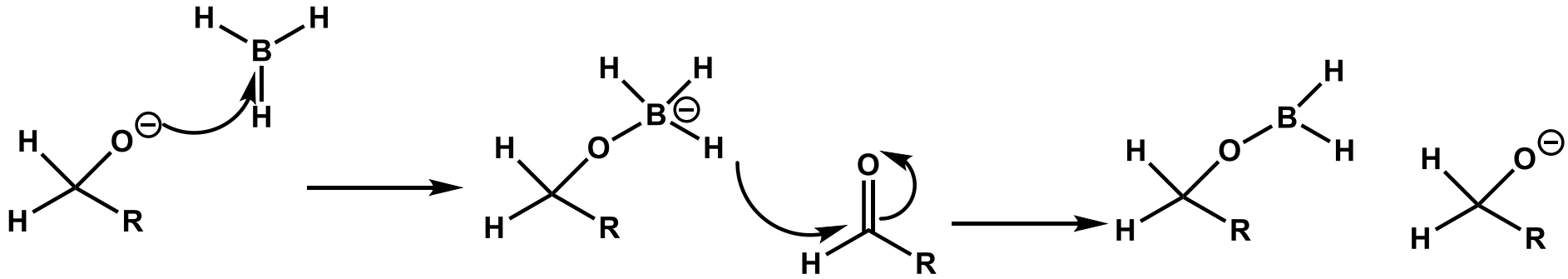
Eight electrons in B-H bonds

Impossible structure: ten electrons in bonds to B

Eight electrons in B-H bonds

Six electrons in B-H bonds and one empty p orbital





Summary

- **Why the C=O group reacts with nucleophiles:**
- ✓ **Polarity of the C=O group**
- ✓ **Molecular orbitals involved**

Summary

- **Why the C=O group reacts with nucleophiles:**
- ✓ **Polarity of the C=O group**
- ✓ **Molecular orbitals involved**
- **What sorts of molecules can be made by reactions of C=O groups**
- ✓ **Cyanohydrines by cyanide addition**
- ✓ **Alcohols by reduction**

Summary

- **Why the C=O group reacts with nucleophiles:**
- ✓ **Polarity of the C=O group**
- ✓ **Molecular orbitals involved**
- **What sorts of molecules can be made by reactions of C=O groups**
- ✓ **Cyanohydrins by cyanide addition**
- ✓ **Alcohols by reduction**
- **Looking forward to:**
- ➢ **C-C bond formation by organometallic reagents**
- ➢ **Diols by water addition**
- ➢ **Hemiacetals by alcohol addition**
- ➢ **How acid and basis catalysis improve reactivity**